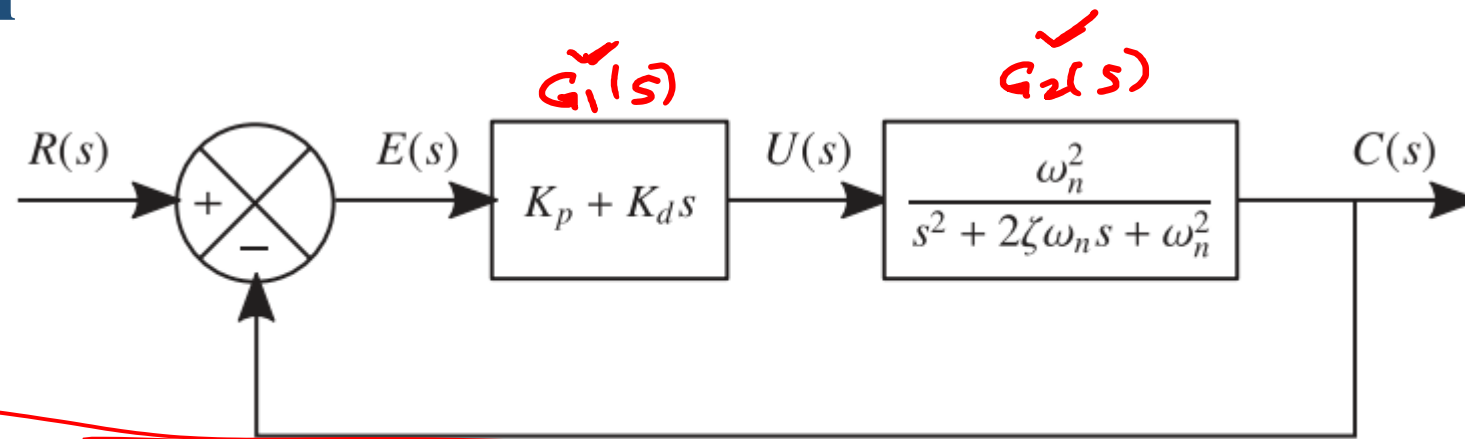


# Proportional Plus Derivative (PD) Control of a Second-Order System



$$\frac{C(s)}{R(s)} = \frac{G_1(s)G_2(s)}{1 + G_1(s)G_2(s)}$$

$$R(s) = \frac{1}{s}$$

$$\frac{C(s)}{R(s)} = \frac{(K_p + K_d s) \omega_n^2}{s^2 + (2\zeta \omega_n + K_d \omega_n^2) s + (1 + K_p) \omega_n^2}$$

$$C_{ss} = \lim_{s \rightarrow 0} s C(s) = \lim_{s \rightarrow 0} s \times \frac{1}{s} \times \frac{(K_p + K_d s) \omega_n^2}{s^2 + (2\zeta \omega_n + K_d \omega_n^2) s + (1 + K_p) \omega_n^2}$$

$$= \frac{K_p \omega_n^2}{(1 + K_p) \omega_n^2} = \frac{K_p}{1 + K_p}$$

$$\frac{E(s)}{R(s)} = \frac{1}{1 + G_1(s)G_2(s)}$$

$$R(s) = \frac{1}{s}$$

$$\frac{E(s)}{R(s)} = \frac{s^2 + 2\xi\omega_n s + \omega_n^2}{s^2 + (2\xi\omega_n + K_d\omega_n^2)s + (1 + K_p)\omega_n^2}$$

$$e_{ss} = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} s \times \frac{1}{s} \times \frac{s^2 + 2\xi\omega_n s + \omega_n^2}{s^2 + (2\xi\omega_n + K_d\omega_n^2)s + (1 + K_p)\omega_n^2}$$

↓ 0

↓ c.

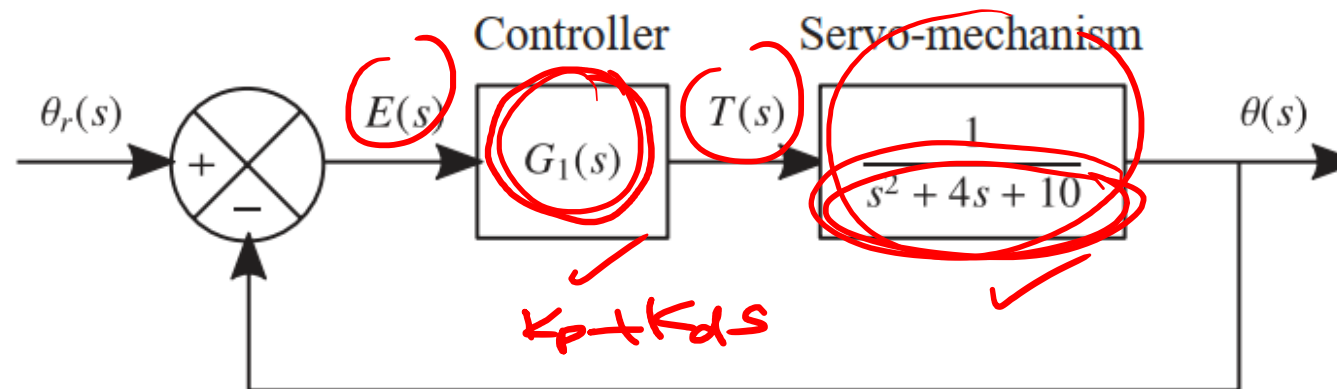
$$= \frac{\cancel{\omega_n^2}}{(1 + K_p)\cancel{\omega_n^2}}$$

$$= \left( \frac{1}{1 + K_p} \right) //$$

# Exercise 1

The angular output of a servo-mechanism  $\theta$  is controlled as indicated in the figure below where  $\theta_r$  is the required (reference) displacement. The controller transfer function is given by  $G_1(s)$ , the torque it applies to the servo-mechanism is denoted by  $T(s)$  and the error signal  $E(s)$ .

- Calculate the damping and natural frequency of the uncontrolled open-loop system. What is the maximum overshoot of the uncontrolled system in response to a unit step change in torque  $T(s)$ ?
- Proportional control with gain  $K_p = 10$  is applied. Calculate the closed-loop natural frequency and damping. What is the steady state error in response to a unit step input in  $\theta_r(s)$ ? Calculate the resulting maximum overshoot.
- Derivative action with gain  $K_d = 4$  is now added to the proportional action. Calculate the closed-loop natural frequency and damping. What is the steady state error in response to a unit step input in  $\theta_r(s)$ ? Calculate the resulting maximum overshoot.



$$O/L \ T/f = G_1(s)G_2(s) = G_1(s) \cdot \frac{1}{s^2 + 4s + 10}$$

$$ch \ e \Rightarrow s^2 + \underline{4}s + \underline{10} = 0$$

$$s^2 + \underline{2\zeta\omega_n}s + \underline{\omega_n^2} = 0$$

$$\omega_n^2 = 10 \Rightarrow \omega_n = \underline{3.162} \text{ rad/s}$$

$$2\zeta\omega_n = 4 \Rightarrow \zeta = \frac{4}{2 \times 3.162} = \underline{0.6325}$$

$$M_p = \underline{100} e^{-\zeta\omega_n t_p} (\%)$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$M_p = \underline{e^{-\left(\frac{\pi\zeta}{\sqrt{1-\zeta^2}}\right)}}$$

$$t_p = \frac{\pi}{\omega_d}$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$t_s = \frac{3}{\zeta\omega_n} = \frac{3}{0.6325 \times 3.162} = \underline{1.5 \text{ s}}$$

$$\omega_d = 2.4492$$

$$M_p = \underline{e^{-\left(\frac{\pi \times 0.6325}{\sqrt{1 - 0.6325^2}}\right)}} = 7.6892\%$$

$$t_p = \frac{\pi}{\omega_d} = 1.2827 \text{ s}$$

b)

$$C/L \ T/f = \frac{G_1(s) G_2(s)}{1 + G_1(s) G_2(s)} = \frac{10 \times \left( \frac{1}{s^2 + 4s + 10} \right)}{1 + 10 \times \left( \frac{1}{s^2 + 4s + 10} \right)} = \frac{10}{s^2 + 4s + 20}.$$

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$s^2 + 4s + 20 = 0$$

$$\omega_n^2 = 20 \Rightarrow \omega_n = 4.472 \text{ rad/s}$$

$$2\zeta\omega_n = 4 \Rightarrow \zeta = \frac{4}{2 \times 4.472} = 0.4472$$

$$\frac{E(s)}{R(s)} = \frac{1}{1 + G_1(s) G_2(s)} = \left( \frac{s^2 + 4s + 10}{s^2 + 4s + 20} \right)$$

$$E(s) = \frac{1}{s} \times \left( \frac{s^2 + 4s + 10}{s^2 + 4s + 20} \right)$$

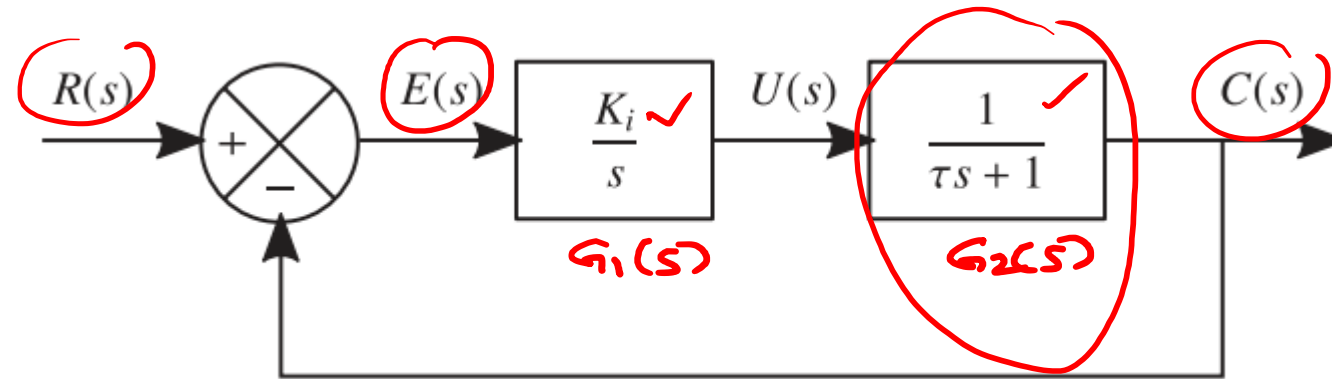
$$R_{ss} = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} s \times \frac{1}{s} \left( \frac{s^2 + 4s + 10}{s^2 + 4s + 20} \right) = \frac{10}{20}$$

$$= 0.5$$

$$M_p = 20.789 \%$$



# Integral Control of a First-Order System



$$\frac{C(s)}{R(s)} = ?$$

$$\frac{E(s)}{R(s)} = ?$$

$$R(s) = 1/s$$

$$\downarrow$$

$$C_{ss} ? \quad E_{ss} ?$$

$$\frac{C(s)}{R(s)} = \frac{G_1 G_2}{1 + G_1 G_2} = \frac{K_i}{\tau s^2 + s + K_i}$$

$$R(s) = 1/s$$

$$C(s) = \frac{1}{s} \left( \frac{K_i}{\tau s^2 + s + K_i} \right)$$

$$C_{ss} = \lim_{s \rightarrow 0} s C(s) = \lim_{s \rightarrow 0} s \times \frac{1}{s} \times \left( \frac{K_i}{\tau s^2 + s + K_i} \right) = \frac{K_i}{K_i}$$

$$= 1$$

$$\frac{E(s)}{R(s)} = \frac{1}{1 + G_1 G_2} = \frac{s(\tau s + 1)}{\tau s^2 + s + K_i}$$

$\downarrow$   
 $1/s$

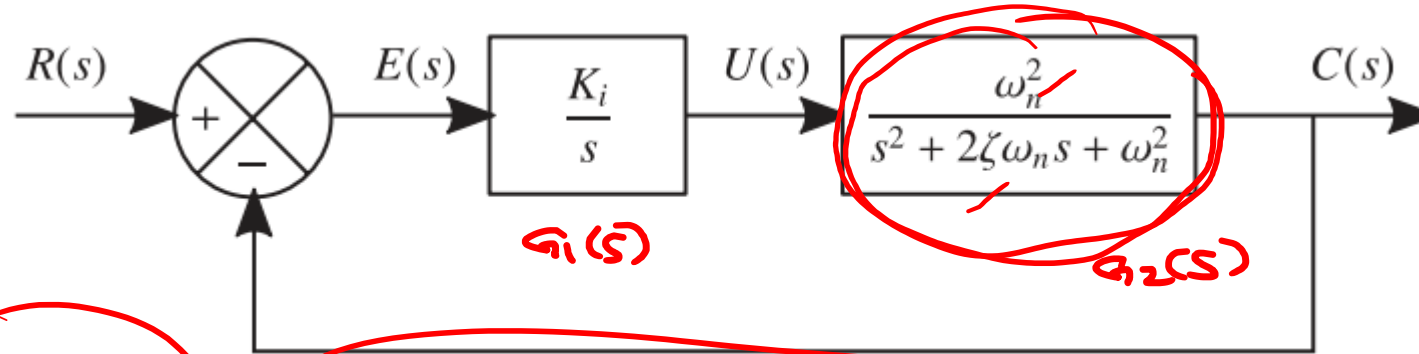
$$E(s) = \frac{1}{s} \frac{s(\tau s + 1)}{\tau s^2 + s + K_i}$$

$$e_{ss} = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} \cancel{s} \times \frac{1}{\cancel{s}} \times \frac{s(\tau s + 1)}{\tau s^2 + s + K_i} = \frac{0}{K_i}$$

$$= 0 \quad \checkmark$$



# Integral Control of a Second-Order System



$$\frac{C(s)}{R(s)} = \frac{G_1 G_2}{1 + G_1 G_2} = \frac{K_i \omega_n^2}{s^3 + 2\zeta \omega_n s^2 + \omega_n^2 s + K_i \omega_n^2}$$

$$R(s) = \frac{1}{s} \Rightarrow C(s) = \frac{1}{s} \times \left( \frac{K_i \omega_n^2}{s^3 + 2\zeta \omega_n s^2 + \omega_n^2 s + K_i \omega_n^2} \right)$$

$$C_{ss} = \lim_{s \rightarrow 0} s C(s) = \lim_{s \rightarrow 0} s \times \frac{1}{s} \times \frac{K_i \omega_n^2}{s^3 + 2\zeta \omega_n s^2 + \omega_n^2 s + K_i \omega_n^2}$$

$$= \frac{K_i \omega_n^2}{K_i \omega_n^2} = 1$$

$$\frac{E(s)}{R(s)} = \frac{1}{1 + G_1 G_2} = \left( \frac{s(s^2 + 2\zeta\omega_n s + \omega_n^2)}{s^3 + 2\zeta\omega_n s^2 + \omega_n^2 s + K_i \omega_n^2} \right)$$

$$R(s) = \frac{1}{s} \Rightarrow E(s) = \frac{1}{s} \times ( )$$

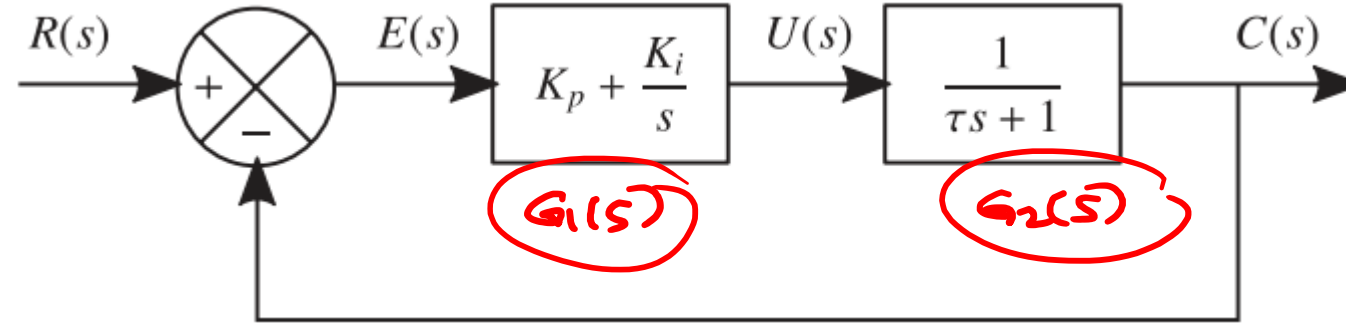
$$P_{ss} = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} s \times \frac{1}{s} \times \frac{s(s^2 + 2\zeta\omega_n s + \omega_n^2)}{s^3 + 2\zeta\omega_n s^2 + \omega_n^2 s + K_i \omega_n^2}$$

$\downarrow$   
 $0$

$$= \frac{0}{K_i \omega_n^2}$$

$$= 0$$

# PI Control of a First-Order System



$$\frac{C(s)}{R(s)} = \frac{G_1 G_2}{1 + G_1 G_2} = \frac{K_p s + K_i}{\tau s^2 + (K_p + 1)s + K_i}$$

$$R(s) = \frac{1}{s}$$

$$C_{ss} = 1$$

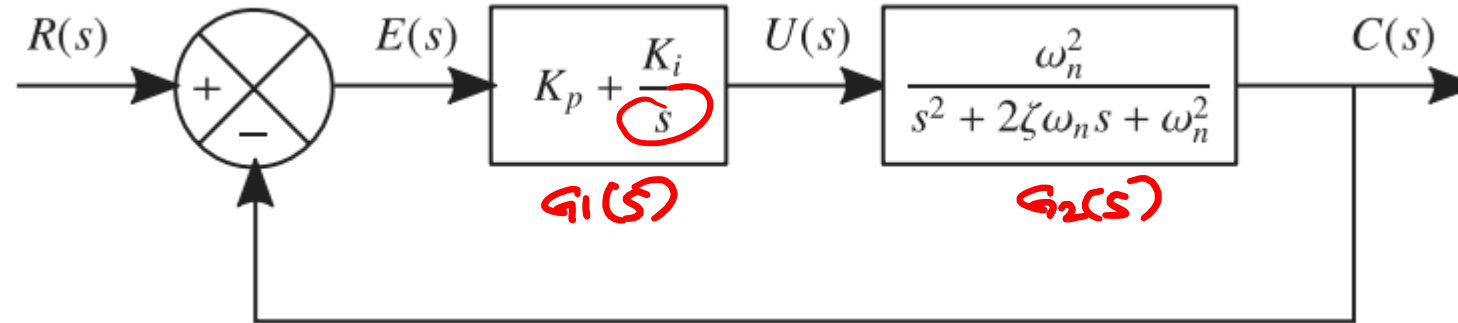
$$\frac{E(s)}{R(s)} = \frac{1}{1 + G_1 G_2} = \frac{s(zs + 1)}{zs^2 + (K_p + 1)s + K_i}$$

$\frac{1}{s}$  ✓

$E(s)$  ✓

$R_{ss} = 0$

# PI Control of a Second-Order System



$$\frac{C(s)}{R(s)} = \frac{(K_i + K_p s) \omega_n^2}{s^3 + 2\zeta \omega_n s^2 + \omega_n^2 (K_p + 1) s + K_i \omega_n^2}$$

$$R(s) = 1/s$$

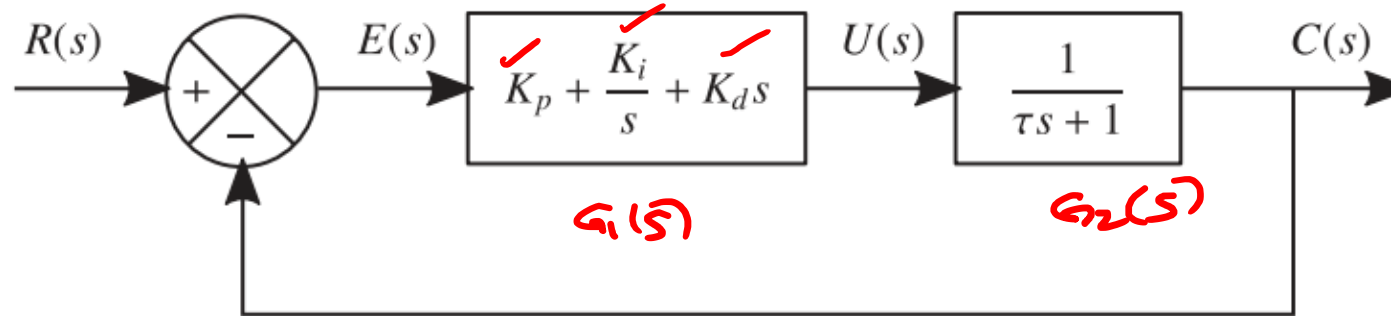
$$C_{ss} = 1$$

$$\frac{E(s)}{R(s)} = \frac{s(s^2 + 2\zeta\omega_n s + \omega_n^2)}{s^3 + 2\zeta\omega_n s^2 + \omega_n^2(K_p + 1)s + K_i\omega_n^2}$$

$\swarrow$   
 $1/s$   
 $E(s)$  ✓

$\underline{\underline{p_{ss} = 0}}$

# PID Control of a First-Order System



$$\frac{C(s)}{R(s)} = \frac{K_d s^2 + K_p s + K_i}{(z + 1)s^2 + (K_p + 1)s + K_i}$$

$\downarrow$   
 $y_s$

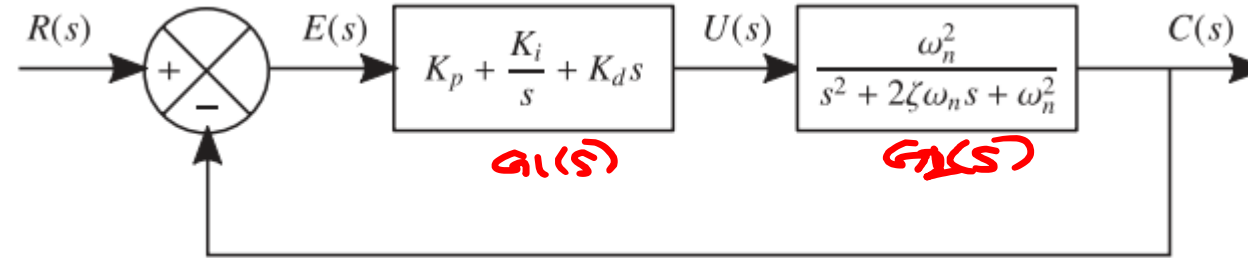
$$C_{ss} = 1$$

$$Y_S \leftarrow \frac{\overline{E(s)}}{\overline{R(s)}} = \frac{s(zs+1)}{(z+k_d)s^2 + (k_p+1)s + 1k_i}$$

$e_{ss} = 0$



# PID Control of a Second-Order System



$$\frac{C(s)}{R(s)} = \frac{(K_i + K_p s + K_d s^2) \omega_n^2}{s^3 + (2 \zeta \omega_n + K_d \omega_n^2) s^2 + \omega_n^2 (K_p + 1) s + K_i \omega_n^2}$$

$$C(s) = R$$

$$C_{ss} = 1$$

# PID Control of a Second-Order System cont.

$$Y_s \leftarrow \frac{E(s)}{R(s)} = \frac{s(s^2 + 2\xi\omega_n s + \omega_n^2)}{s^3 + (2\xi\omega_n + K_d\omega_n^2)s^2 + \omega_n^2(K_p + 1)s + K_i\omega_n^2}$$

$E(s)$  ✓

$$\underline{\underline{e_{ss} = 0}}$$